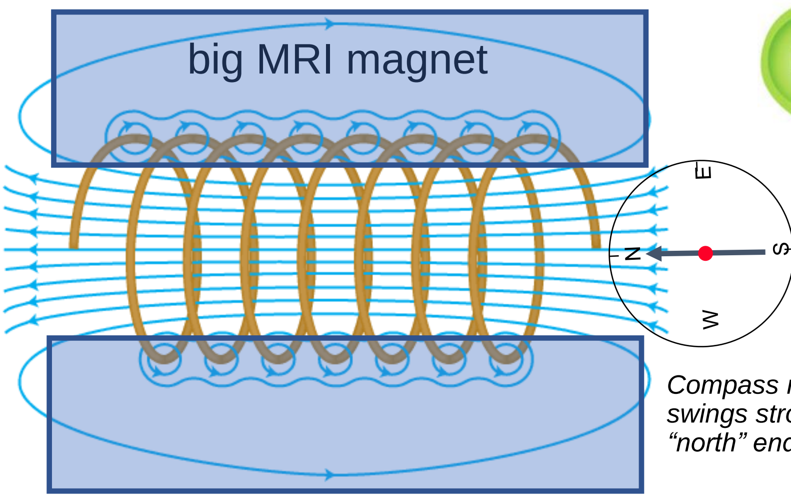
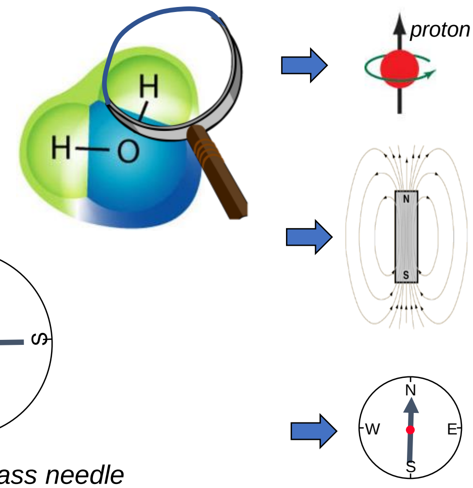
Protons (little magnets) in water do the same in big MRI magnet





Compass needle swings strongly toward "north" end of magnet

Technical difference: proton is not constrained by pivot to 2-dimensional motion…

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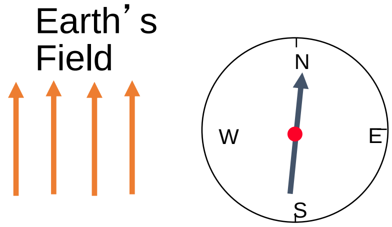
Magnetic moment in an external B field

Earth's Field

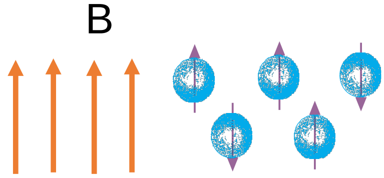


compass

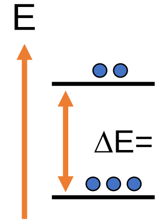
Two ways of saying the same thing:

1) The field torques the needle (magnetization) into alignment.
2) Alignment along north is the lowest energy state: E = -cos θ.

B



protons



Weird twist thanks to quantum mechanics:

Only two observable energy states: "up" and "down"

But we only look at ensembles
Total E: add… practically a continuous variable…

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